Structure and randomness in the prime numbers (IMO Festschrift submission)

Terence Tao · 14 September 2009

https://terrytao.wordpress.com/2009/09/14/structure-and-randomness-in-the-prime-numbers-imo-festschrift-submission/

tao2typ · converted to Typst 2026-10-14 (UTC)

A few months ago, I gave a talk at the IMO in Bremen on "Structure and randomness in the prime numbers". I have now converted the slides from that talk into a more traditional paper (7 pages in length), for submission to a Festschrift for the Bremen Olympiad. The content is much the same as the slides, but some references have been added.

Comments (9)

 **Anonymous** · 14 September 2009

Interesting! (There's a typo in the third paragraph of the first section)

 **Manjil P. Saikia** · 15 September 2009

Interesting, maybe the other talks like the one on Poincare Conjecture can also be made into such a format.

 **Thuris** · 16 September 2009

Thank you for posting this. Although a non-mathematician, I found it very clear. (Which gives me hope to understand some of the other posts eventually!)

 **Hassan Mohy-ud-Din** · 17 September 2009

Greetings Professor Terence Tao.
How be you? I am a PhD student at Johns Hopkins University. My request is that if you can kindly upload the videos of the talks/lectures along with the slides it will be very helpful and also make a tab of upcoming seminars/conferences/workshops. Thank You
Good Bye

 **xi'an** · 17 September 2009

Dear Terence, there is a typo in the references: de La Vallée Poussin's book is "Recherches analytiques sur la théorie des nombres premiers"
Best regards,
Xi'an

 **zeynel** · 30 September 2009

Yes, thank you for the slides. I enjoyed reading it as a non-mathematician. In your last slide you say that primes are *mostly* odd. Does this suggest that there may be even primes except 2?

 **ARN** · 30 September 2009

Zeynel , I think that was probably a joke. There can be no even primes > 2.

links for 2009-09-30 « Blarney Fellow · 30 September 2009
[...] Structure and randomness in the prime numbers (IMO Festschrift submission) « What's new (tags: random primes math complexity computation) [...]

 **Tom Verhoeff** · 26 January 2011

The grammar seems broken in the sentence containing "to not obey behave pseudorandomly". You can delete "obey" to fix the grammar, but probably "not" should be deleted as well to fix the semantics. This is both on the slides and in the paper.
[Thanks! I think this was fixed in the published version of the text – T.]